

ABSTRACT

In July of 2020, my dad helped me buy a 2011 Subaru Impreza with 64k miles for \$6k. On December 6th, 2021, I got in a car accident where my insurance deemed my car a total loss. My Impreza had 87k miles & I received \$7.2k. To give you a look at the current used car market: If I wanted to buy the same Impreza, I bought **2 years ago** with the same mileage, it would cost me \$8.8-\$11.5k (that's a **46%-91% price increase**). This time around, I must **search** and **buy** a car myself. Not knowing anything about cars, I decided it would be a great idea to use data science to help me learn about the things associate with high car prices. Having a budget of 15k, this is going to help me look avoid certain things.

INTRODUCTION

Using Kaggle I acquire the ‘100k UK Used Car Data set’ to conduct this analysis. From this data set, we will focus on 6 variables: Engine Size, Year, Mileage, Transmission, Make, and Fuel Type. These variables will have an overlay of my budget to help visualize what associates with a car costing more than \$15k. This will help direct my focus when looking for a car.

REFERENCES

“2011 Subaru Impreza Values & Cars For Sale | Kelley Blue Book.
<https://www.kbb.com/subaru/impreza/2011/>.
Aditya. “100,000 UK Used Car Data Set.” *Kaggle*, 4 July 2020,
<https://www.kaggle.com/adityadesai13/used-car-dataset-ford-and-mercedes>.

EXPLORATORY DATA ANALYSIS

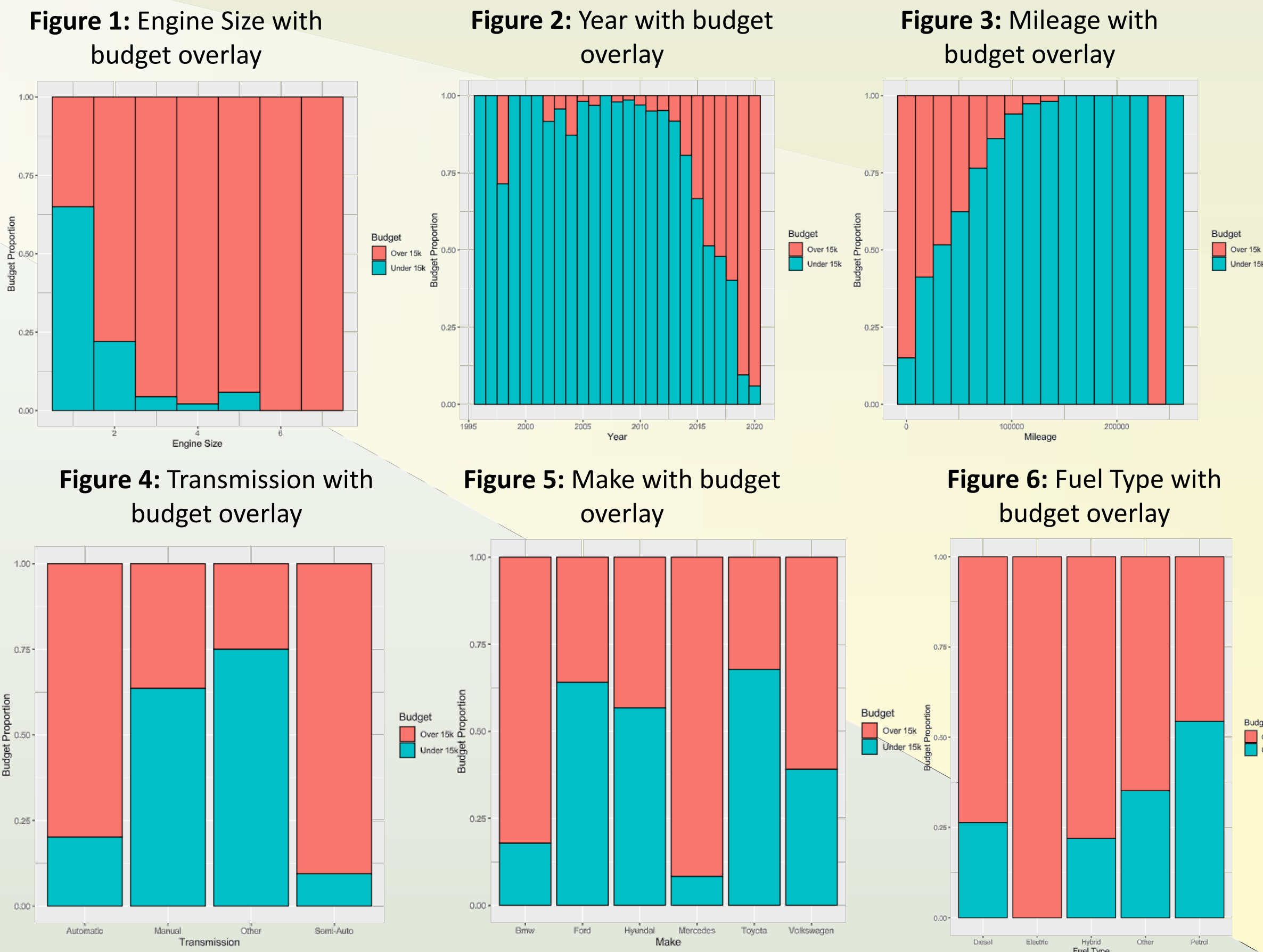


Figure 1 shows that as the engine size increases, the cost of the car increases as well. **Figure 2** shows newer cars have a smaller proportion of costing <\$15K. More than 90% of cars built in 2019-20 cost >\$15k. **Figure 3** shows that as the mileage of a car increases, the higher the proportion of it costing less than \$15K increases. **Figure 4** Shows that Semi-Auto & Automatic cars are more expensive than Manual or *Other (Other refers to **Continuously variable transmission (CVT)**) **Figure 5** shows that Toyota, Fords & Mercedes are the brands that have the highest proportion of cars costing <\$15K. All of them have a proportion higher than .58. **Figure 6** shows that most the Fuel Type associated most less expensive cars is Petrol. All Electric Vehicles are more than \$15k.

Model & Results

METHODOLOGY
To obtain our results, 10-fold cross validation was used on CART decision tree models.

This model results in 3 terminal nodes classifying 59% of our data into the “Over \$15K” category and 41% into the “Under \$15K” category.

Our first node indicates that 60% of the cars in our data set cost more than \$15k. The best way to find a car under \$15k according to our CART Model is to look for a car that has a manual Transmission and was made before 2019.

Decision Tree Structure:

- Node 1: **TransmissionManual < 0.5**
 - yes (60%): Node 2: **Over 15k** (.60 .40, 100%)
 - no (40%): Node 3: **Under 15k** (.36 .64, 53%)
 - Node 4: **Year >= 2019**
 - yes (25% of total): Node 6: **Over 15k** (.77 .23, 12%)
 - no (75% of total): Node 7: **Under 15k** (.25 .75, 41%)

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CONCLUSION

Our model shows the most strength when it identifies cars that cost more than our budget of \$15,000. It correctly Identifies these cars 83.44% of the time. Additionally, this model has a high specificity of 83% when identifying cars over \$15,000. This is useful to know because when deciding on factors to buy a car, I would want more accurate information on what makes a car more expensive so I can avoid these things. This saves me time by giving me a smaller list to look for in cars. The Overall accuracy of our entire model is 80%, which is good considering that 60% of the cars in this entire data set are over our budget. It is apparent that 2 attributes that impact a cars high price is: The type of Transmission and the Year. This is going to reduce the number of cars I look for when I look for a Honda’s with less than 100,000 miles.